

Exploring Prices and Trading Volumes of SPX and DJ

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Motivation and Research Questions:

Indexes seem to be a good asset to increase wealth over time, reducing the exposure to risk by diversifying. The USA stock market has proven to be one of the best markets to invest in given its return and given the fact that it is nowadays one of the biggest and most efficient market in the world considering the amount of private investors and financial institutions that invest in it. The most popular index is probably the S&P500. This is just one of many indexes that are now available; other important indexes are the Nasdaq index and the Dow Jones index. One of the main questions is whether these indexes share common characteristics, such as similarities in price dynamics, trading volume, and overall market behavior. The aim of this paper is to compare two indexes, the S&P500 and the Dow Jones to assess whether they have similar behaviour, and to compare them; these two indexes, in particular are interesting because they are constructed differently; the S&P500 is a market cap index and it contains 500 firms, while the Dow Jones is a price-weighted and it includes 30 big firms (blue chips), usually leaders in their respective sectors. This difference may lead to differences in market reactions, volatility and trading volume; the conclusions of this paper could find many real world applications especially in portfolio management decisions.

Literature review:

The efficient market hypothesis introduced by Fama² showed that market price movements are unpredictable and exhibit a random walk behaviour. Other researchers, such as Turner & Weigel reveal that daily volatility is persistent and significant in US equity markets, while Hansen & Lunde demonstrate that GARCH (1,1) models capture dynamics more effectively than ARCH. Fleming & Kirby showed that volatility and volumes exhibit persistence (long memory), while Gil-Alana, Infante & Martín-Valmayor showed that SPX and DJIA don't show any cointegration using daily data. This paper aims to review these findings and to get some insights about the model relations of these two indexes.

Data:

The excel file used in this analysis is constructed as it follows:

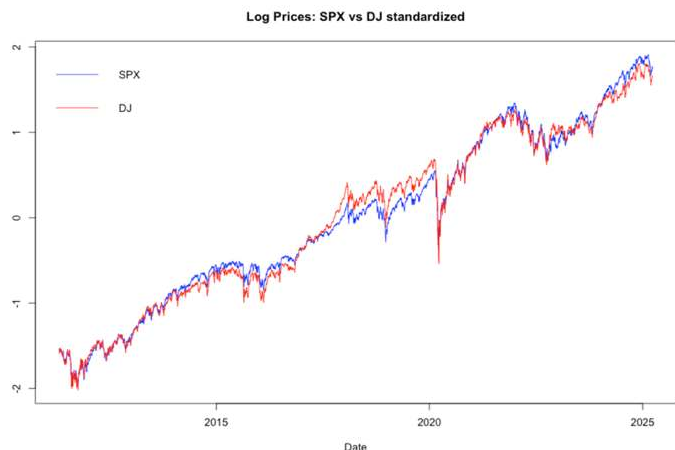
Column	Content
A	Date (daily)
B	Daily S&P 500 price
C	Daily trading volume of S&P 500
D	Daily Dow Jones price
E	Daily trading volume of Dow Jones

Closing price and volume traded for both indexes were collected using Bloomberg through CBS terminals, covering the period from April 2011 to March 2025. This time interval has been selected primarily for two reasons; the analysis uses trading volume that has become more available and consistent in recent years; the selected period contains 3500 observations containing different economic phases (such as expansions, contractions and shocks).

Analysis with ARIMA:

We start by constructing the database as explained above; some changes in the data import are required, including skipping a row, arranging the data in ascending order and other adjustments. Note: the original dataset has the prices in levels; by using R we can transform them into log prices; this has better applications because by working with log prices we can look at the daily % change (same for trading volumes). We can immediately plot the graphs of the log prices of the S&P500 and the log prices of the Dow Jones. The two graphs look very similar, however we can't directly compare them because the prices are not in scale. To directly compare them, we can standardized price data of both indexes and, finally, plot them to check if we can get some insights.

Despite the fact that these two indexes are built differently, they show similar behaviour, showing that both stock prices and market capitalization offer key market insights. By looking at the graph, however, we can see that the Dow Jones has shown higher peaks and deeper downturns. This behaviour is justified by the number of firms included in the two indexes (500 firms vs 30 firms). Having fewer firms in the index makes it less diversified and consequently more exposed to the market.



We can see that there is a trend in both indexes; to further analyze this time series we should remove time trends and make the series stationary. We can test this by applying the ADF test (type=trend) and the KPSS test to assess whether the series are stationary or not. The result suggests the presence of a time trend significant at the 5% level; our model is the following:

$$\text{Model: } \log P(t) = a_0 + a_1 t + \log P(t-1) + \varepsilon^t \quad \text{Moments: } E(y_t) = y_0 + a_0 t \\ \text{Var}(y_t) = t \sigma^2$$

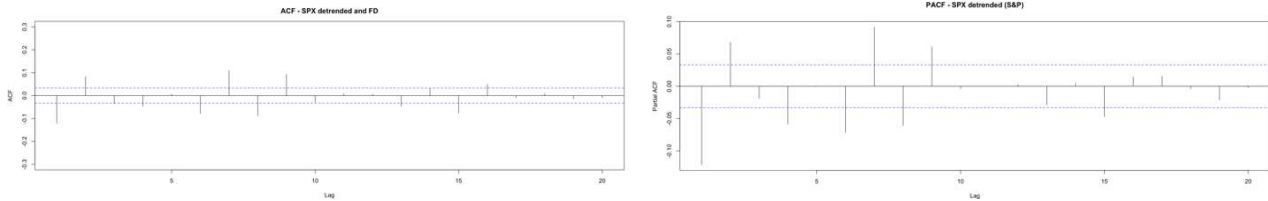
At this point, in order to model the series properly, we need to detrend them:

$$\text{Model: } \log P(t)_{\text{detrended}} = \log P(t) - \hat{a}_0 - \hat{a}_1 t \quad \text{Moments: } E(y_t) = 0 \\ \log P(t)_{\text{detrended}} = b \log P(t-1) + u^t \quad \text{Var}(y_t) = \sigma^2$$

After we treated the two time series by detrending them we can observe the ACF and the PACF functions. In both series, we can see a slow, linear, decaying pattern in the ACF, while the PACF function becomes insignificant after the 10th LAG (still, it has some significant values afterwards); we should probably take into account the presence of a stochastic component. By plotting the graph of the detrended model, we can notice some fluctuations that suggest the presence of a random component. We need to differentiate the two series.

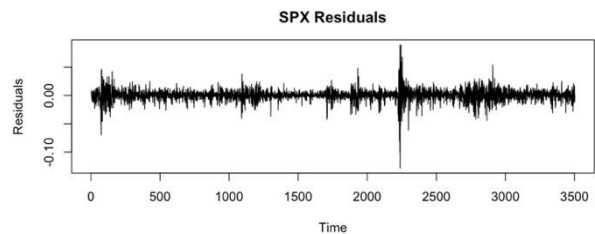
We get the following model: $\Delta \log P(t)_{\text{detrended}} = u^t$

After detrending and first differencing we get the following ACF and PACF functions:



Note: only the S&P500 ACF and PACF function are shown because the ACF and PACF function of the DJ are very similar (same for the residual image below). There are some differences about the length of the bars, however we can say that most of the significant and insignificant Lags are the same for both indexes. The conclusion is that both indexes have a stochastic and a deterministic component and that they are probably going to be modeled similarly. We can check now if the residuals behave like white noise by running the Ljung test. We get a very small p-value, so we reject the null hypothesis of well-behaving residuals. Further modelling is required.

We can start to construct our ARMA model. Notice that in the code I have used `auto.arima()` (to double check whether integration is required). Our model will not include a drift term because we have eliminated it by detrending the series and we expect the series to not be integrated. The last significant



Lag in the ACF is at lag 16, while for the PACF function, the last significant Lag is Lag 15 with many significant Lags in both the ACF and PACF (probably spurious correlations); it's really difficult to understand what type of model should be used. For this reason we use `auto.arima()` function to estimate the model. We estimate the model by trying to minimize the AIC and the BIC model for both indexes. By minimizing the AIC and the BIC we get that the best model is the Arima (2,0,2) for both indexes.

This is good news, because this means that the two indexes can be modelled in the same way, which suggests that the two series are very similar. The models we get are:

$$\begin{array}{llllll} \text{Model S\&P500:} & \log P(t) = & -1.7107 & P(t-1) & -0.8636 & P(t-2) & +1.6129 & \varepsilon(t-1) & +0.7621 & \varepsilon(t-2) & + \varepsilon t \\ & \text{s.e.} & & (0.0266) & & (0.0252) & & (0.0336) & & (0.0319) \end{array}$$

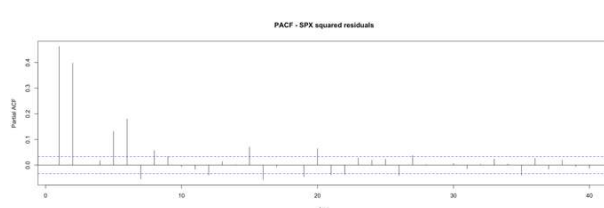
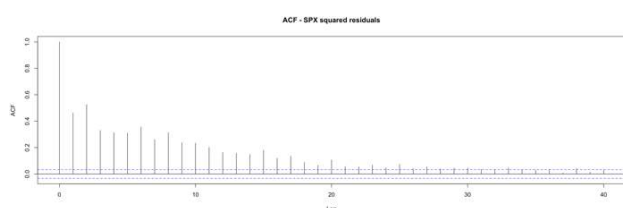
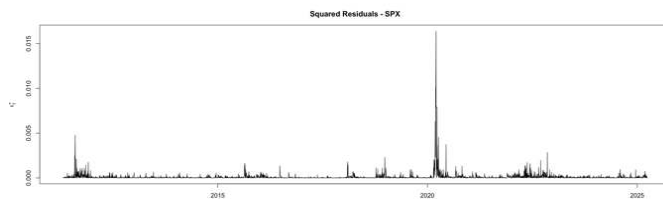
$$\begin{array}{llllll} \text{Model DJ:} & \log P(t) = & -1.7155 & P(t-1) & -0.8670 & P(t-2) & +1.6114 & \varepsilon(t-1) & +0.7543 & \varepsilon(t-2) & + \varepsilon t \\ & \text{s.e.} & & (0.0238) & & (0.0225) & & (0.0310) & & (0.0290) \end{array}$$

Given that all the coefficients are very similar one to another across Lags, that they are all statistically significant and that the residuals behave like white noise up to Lag 4, we can conclude that both indexes are driven by common mechanics. This result is confirmed by running the Ljung Box test.

Volatility analysis with GARCH:

To model volatility, we start by looking at the square residual plots. We can see two abnormal peaks, one in the beginning of the sample (2011-2012) and one in 2020 that is much higher and looks abnormal. The first

one is probably due to the debt crisis in Europe and the US grade rating downgrade, while the second one is clearly due to the market reaction to the pandemic of Covid-19. Having an idea of how the volatility changed over time, we can start to take a look at the ACF and PACF functions. Below we can see the S&P500 ACF and PACF (very similar to DJ ACF and PACF):



We can see that the Lags are autocorrelated for both indexes, this is the type of pattern that we expect from time-dependent (heteroschedastic) series. This means that the variance is not constant over time, but rather time-dependent, which is a violation of the classical assumption of homoscedasticity. For this reasons we can model the volatility. By using the ARCH test we see that the ARCH model could actually be appropriate to model the volatility because the pvalue that we get is very small, $2.2e-16$ in both series. By looking at the ACF functions we see significant values over Lag 30, so a GARCH model would be probably a better choice for modelling. At this point it's necessary make some assumptions. We are going to estimate ARCH/GARCH models by using the maximum likelihood, and, therefore, we need to make assumptions about the distribution of squared residuals. Usually a normal distribution is used, however when modelling with log prices other distributions may be better; during the pandemic period in 2020, for example, Covid has caused abnormal volatility, and this could potentially be an important element to consider in our model. We will start our estimation by assuming a normal distribution and then decide whether another type of distribution should be used. In the code, we can see an attempt of modelling using ARCH (1) and ARCH (2) for both indexes. However, in all those estimations, we can see the following warning messages: "Warning:singular information", "* FALSE CONVERGENCE *" or NA coefficients. ARCH model doesn't capture the conditional variance, suggesting that all the estimates are not reliable; for this reason, an estimation with GARCH model seems reasonable. I decided to estimate an ARMA (2,2)+GARCH (p,q) model (rather than only GARCH) because it's necessary to take into consideration the autocorrelation in the mean and to make sure residuals exhibit a good behaviour. To decide what p,q to choose I am going to compare AICs and BICs values and compare them while modelling with the most common GARCHs specifications, such as GARCH (1,1), GARCH (2,1), GARCH (1,2) and GARCH (2,2). By running all these models we see that a normal distribution is not representative of the distribution of the residuals because we get Jarque-Bera p-value & Shapiro-Wilk p-value is close to 0. A t distribution is more representative than the normal distribution and

that is what we are going to assume. By comparing the four models it appears that AIC and BIC are very similar in all the GARCH specifications. Moreover, the AIC and BIC show lower values for the t distribution models with respect to the normal distribution models. For these reasons we are going to model using ARMA (2,2)+GARCH (1,1) and by assuming a t distribution. The results we get are the following:

Param	S&P500					DJ			
	Estimate	Std Error	t value	Pr(> t)		Estimate	Std Error	t value	Pr(> t)
mu	2.569e-04	2.148e-04	1.196	0.232		4.437e-04	3.397e-04	1.306	0.192
ar1	4.320e-01	8.606e-01	0.502	0.616		2.136e-01	4.036e-01	0.529	0.597
ar2	3.426e-01	6.978e-01	0.491	0.623		3.483e-01	3.021e-01	1.153	0.249
ma1	-4.874e-01	8.565e-01	-0.569	0.569		-2.476e-01	4.069e-01	-0.609	0.543
ma2	-3.462e-01	7.317e-01	-0.473	0.636		-3.432e-01	3.142e-01	-1.093	0.275
omega	1.195e-05	2.568e-06	4.655	3.24e-06 ***		1.896e-05	3.748e-06	5.059	4.21e-07 ***
alpha1	1.660e-01	1.883e-02	8.816	<2e-16 ***		1.657e-01	1.941e-02	8.541	<2e-16 ***
beta1	8.256e-01	1.745e-02	47.310	<2e-16 ***		8.154e-01	1.900e-02	42.913	<2e-16 ***
shape	5.662e+00	5.440e-01	10.408	<2e-16 ***		5.904e+00	5.960e-01	9.906	<2e-16 ***

Test	Series	Stat Type	Statistic	p-Value	Statistic	p-Value
Jarque-Bera Test	R	Chi^2	1005,701	0.000	601,49	0.000
Shapiro-Wilk Test	R	W	0,970913	0.000	0,9792	0.000
Ljung-Box Test	R	Q(10)	11,3537	0.330	10,908	0.364
Ljung-Box Test	R	Q(15)	17,58499	0.285	17,072	0.314
Ljung-Box Test	R	Q(20)	22,4962	0.314	20,905	0.402
Ljung-Box Test	R^2	Q(10)	9,681963	0.4688	11,568	0.314
Ljung-Box Test	R^2	Q(15)	15,23906	0.434	14,23	0.508
Ljung-Box Test	R^2	Q(20)	18,65748	0.544	16,443	0.688
LM Arch Test	R	TR^2	11,91556	0.452	11,629	0.475

By comparing the two series we see a very similar estimation. The coefficients of the ARMA model are very different, however they are not statistically significant. All the coefficients of omega, alpha1, beta1 and shape are statistically significant. Jarque-Bera test and Shapiro-Wilk tests confirm that the series are not normal, while all the tests on autocorrelation reveal no autocorrelation in residuals and in square residuals. The resulting theoretical models are:

S&P500 model (ARMA(2,2) + GARCH(1,1) with t-distribution):
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$$\Delta \log(\text{SPX}_t) = 0.00026 + 0.4320 \cdot \Delta \log(\text{SPX}_{t-1}) + 0.3426 \cdot \Delta \log(\text{SPX}_{t-2}) - 0.4874 \cdot \varepsilon_{t-1} - 0.3462 \cdot \varepsilon_{t-2} + \varepsilon_t$$

$$\sigma_t^2 = 1.195e-05 + 0.1660 \cdot \varepsilon_{t-1}^2 + 0.8256 \cdot \sigma_{t-1}^2$$

$$\varepsilon_t \sim t(5.662)$$

Dow Jones model (ARMA(2,2) + GARCH(1,1) with t-distribution):

$$\Delta \log(\text{DJ}_t) = 0.00044 + 0.2136 \cdot \Delta \log(\text{DJ}_{t-1}) + 0.34839 \cdot \Delta \log(\text{DJ}_{t-2}) - 0.2476 \cdot \varepsilon_{t-1} - 0.3432 \cdot \varepsilon_{t-2} + \varepsilon_t$$

$$\sigma_t^2 = 1.896e-05 + 0.1657 \cdot \varepsilon_{t-1}^2 + 0.8154 \cdot \sigma_{t-1}^2$$

$$\varepsilon_t \sim t(5.904)$$

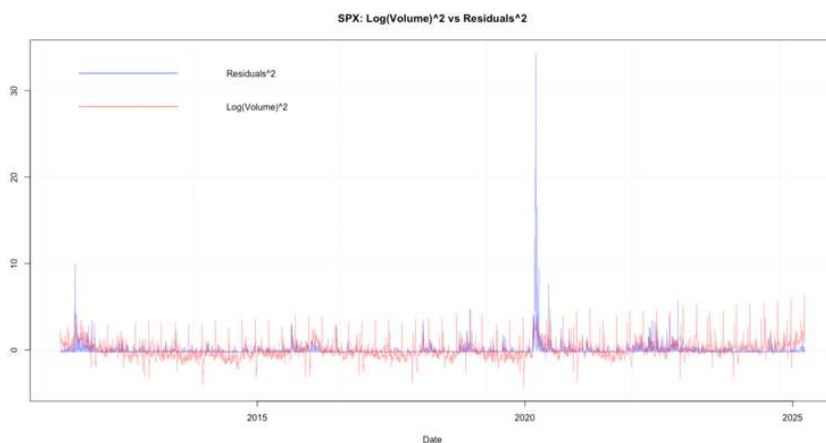
All volatility parameters are significant at the 1% level, ARMA coefficients are not significant. GARCH (1,1) has slightly better AIC and BIC values, but nevertheless results are almost identical. In conclusion, our models show very similar coefficients and behave well. It's interesting to notice that there is an important difference between the two variances of in the long term. To show this, it's sufficient to compute the unconditional variance:

$$\text{Var_SPX} = 1.1953 \cdot 10^{-5} / (1 - 0.16603 - 0.82560) = 1.1953 \cdot 10^{-5} / 0.00837 \approx 0.00143$$

$$\text{Var_DJ} = 1.8960 \cdot 10^{-5} / (1 - 0.16573 - 0.81537) = 1.8960 \cdot 10^{-5} / 0.01890 \approx 0.00100$$

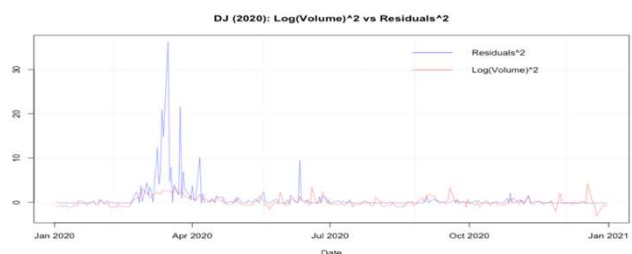
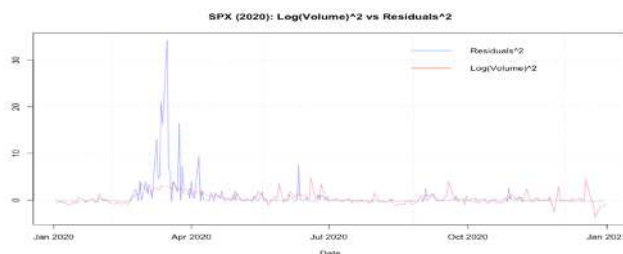
Please notice that volatility modelling was conducted on standardized series for comparability; as a result estimates of long-run variance obtained from GARCH models reflect relative, not absolute, volatility. This suggests that, on average, the relative volatility of the DJ is lower. This is not, however, an unusual result: by looking at the literature, we find other examples where this has been studied. In some papers² authors have shown that, during periods of market stress, the Dow Jones exhibits lower volatility compared to the S&P500 (larger firms are less sensitive to market shocks).

Why is that the case? Probably for multiple reasons: One could be that the S&P500 takes more time to go back to the mean after crises, while big firms are quicker and this causes the volatility to persist for less time. Secondly, maybe smaller firms are more volatile and their volatility affects the sample for a longer period. Third, in crisis periods the high correlation among companies in the S&P 500 amplifies index movements, contributing to elevated volatility. The covid pandemic has surely a big impact on this result. How about the volume? Does it show a strong relationship with the volatility?



By looking at the graph on the left (S&P500) the answer is clearly yes. We can see that during the covid-19 crisis and the sovereign debt crisis of 2011 the volatility has increased (blue line), and so has the traded volumes (red line)³. Moreover, during periods where volatility is low, the traded volume is much lower (on average looks

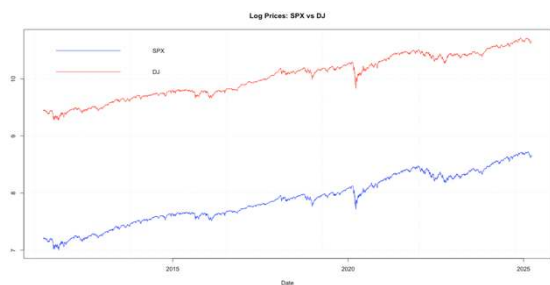
close to 0). To show this we could potentially model the trading volumes similarly to how we have done so for the log prices. The graphs of the two indexes are almost indistinguishable, for this reason I have plotted the residuals squared and the trading volume during the year 2020.



By looking at these graphs we can get two important insights:

- Volatility is likely to increase in crisis periods
- Volumes in the two time series are very similar

Cointegration analysis with VAR/VECM:



We have already shown that the two series must be first differenced to become stationary, now we should use VAR/VECM models in order to get deeper insights into the relationships between these two time series. The graph of the log levels on the left shows clearly that the two series show a similar behaviour and this supports the cointegration analysis that follows.

The idea is to run some tests to understand which model works better:

- The interactions between log prices: useful to check how the prices are related (Granger causality)
- The interactions between log volumes: same concept but applied to traded volumes

The main assumption about the VAR model is that all variables are treated as endogenous. This makes perfectly sense in our context because is very likely that log prices of one index affects the log prices of the other and viceversa. We can proceed by running the Johansen test of log prices. It's important to note that the test is run without taking the first difference, as the test requires. This way I am sure that the cointegration test is reliable. The result is the following:

	test	10pct	5 pct	1 pct	
$r \leq 1$	4.48	10.49	12.25	16.26	insignificant at the 10%
$r = 0$	22.39	22.76	25.32	30.45	insignificant at the 10%

It looks like the log-prices of the two indexes don't share long-run relation 4, not even at the 10% level. To double check this we run another test, the Granger Causality Test. While the Johansen test is useful to determine if there is cointegration in the long term, this test is useful to test the relationship in the short term; the two concepts are related but independent one another. In order to run the test I first differenced both the time series to make the them stationary as the test requires. You will see four outputs: in output I & III am going to test whether the S&P500 helps predicting the DJ, in output II & IV I am going to check if the DJ helps predicting the S&P500 (same for volumes in page 8). The results are:

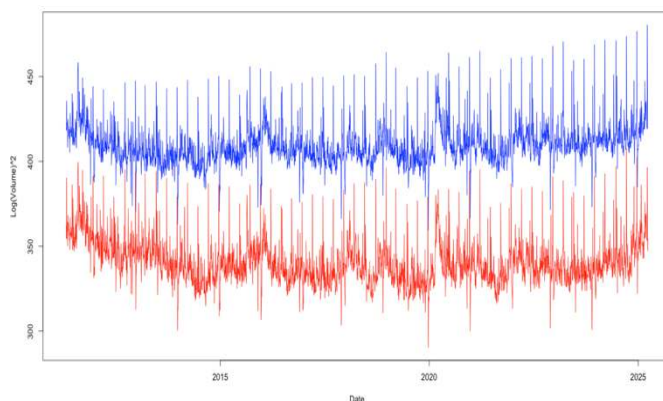
output 1: $F=1.0097$ $pvalue=0.3644$ (not significant)	Using two lags
output 2: $F=4.8803$ $pvalue=0.007647$ (significant at the 0,1% level)	
output 3: $F=1.2041$ $pvalue=0.3009$ (not significant)	Using six lags
output 4: $F=3.3093$ $pvalue=0.002993$ (significant at the 0,1% level)	

This means that the S&P500 prices are not helpful in predicting future DJ prices, however, the DJ prices are helpful in predicting future S&P500 prices according to the test with significant at 0,1%. The direction of the relationship is unidirectional.

Why is the relationship unidirectional? If the DJ prices explain the SPX index and not the opposite, why can't we look at the DJ behaviour and then trade on the SPX to make profits? Why are the two series not cointegrated according to the Johansen test?

Starting from the first question that has a more intuitive answer: it could be the case that DJ firms react more quickly than SPX firms; in other words, apparently, the Dow Jones index is more reactive in the short term but more stable in the long term. This result could be explained by the fact that the 30 firms that are in DJ (that incorporates information quickly) are also in SPX (slower to incorporate information compared to DJ) and this may cause a unidirectional relationship. The answer to the second question is no. According to the market efficiency theory, it's not possible to profit in this way in efficient markets, and in any case there is no economic theory that would support the idea that DJ prices reacts before the SPX prices. We don't have any information about how strong this relationship actually is, if it's stable over time and how long it lasts over time. However, this result is helpful to understand how to model the two time series. The third question is the one that I personally find more interesting and difficult to answer; the two series appear to be $I(1)$ but not cointegrated, however the Granger test suggests that the relationship is only unidirectional. The absence of cointegration could mean that any predictive relationship observed through Granger causality may not be stable or persistent in the long run.

Unlike the ambiguous relationship between prices trading volumes show a clearer directional structure, which likely reflects real market behavior:



In this case it appears that log volumes of S&P500 (blue) are usually higher than log volumes of the Dow Jones index (red). We can see some regularities in the peaks of both time series suggesting the presence of seasonalities. It looks like there is no trend in both series, suggesting that FD is not needed to make the two series stationary.

ADF test suggests that the two time series are already stationary (taking into account the presence of a deterministic trend) for both indexes. There is no need to first differencing the two time series and, therefore, Johansen test cannot be applied here. We can still look at the Granger test. Again, I have run the test twice, the first time with two lags, the second time with six lags. The result is the following:

output 1: $F=31.976$ $pvalue=1.733e-14$ (significant at the 0% level)
 output 2: $F=0.6467$ $pvalue=0.5238$ (not significant)
 output 3: $F=8.9475$ $pvalue=1.018e-09$ (significant at the 0% level)
 output 4: $F=3.1693$ $pvalue=0.004206$ (significant at the 0,1% level)

Using two lags

Using six lags

It looks like the S&P500 trading volumes help predicting the DJ trading volumes but not the opposite when considering two Lags; if we consider six Lags both indexes have explanatory power on the other index. The two Lag output could be explained by the fact that big financial institutions and investors trade primarily on the S&P500, anticipating the movements of the Dow Jones. Another cause could be the fact that S&P500 is used as a macroeconomic and financial instrument and is much more frequently traded.

Now we need to estimate the final model; the question is which model the best model in this case? In this case, the answer is pretty easy: a VECM model is efficient when one assumption is satisfied: The models need to be I(1) and cointegrated: neither the log prices or the log volumes satisfy this assumption; the log prices are I(1) but they are not cointegrated, while the log volumes are I(0). For this reason, we are just going to estimate a VAR model:

VAR OUTPUT

S&P500 log prices differenced significant values			
Variable	Estimate	Residual Std. Error	Significance
DJ_log_diff.l2	0,1097	0,05152 *	
SPX_log_diff.l3	-0,1434	0,0577 *	
DJ_log_diff.l3	0,1174	0,05141 *	
DJ_log_diff.l6	-0,09541	0,05135 .	
DJ_log_diff.l9	0,09276	0,05138 .	
			R-squared
			0,04929
			F-statistic
			10

S&P500 log volumes significant values			
Variable	Estimate	Residual Std. Error	Significance
SPX_log_vol.l1	0,39517	0,2296 ***	
SPX_log_vol.l2	0,20599	0,2296 ***	
DJ_log_vol.l3	0,13536	0,2296 ***	
DJ_log_vol.l5	0,12836	0,2296 **	
SPX_log_vol.l6	0,14008	0,2296 **	
DJ_log_vol.l6	-0,12854	0,2296 **	
const	4,15561	0,2296 ***	
			R-squared
			0,3297
			F-statistic
			85,36

DJ log prices differenced significant values			
Variable	Estimate	Residual Std. Error	Significance
DJ_log_diff.l2	0,1313	0,05774 *	
SPX_log_diff.l3	-0,124	0,06466 *	
DJ_log_diff.l3	0,1002	0,05761 .	
DJ_log_diff.l6	-0,14	0,05754 *	
DJ_log_diff.l9	0,1197	0,05758 *	
			R-squared
			0,05596
			F-statistic
			11,44

DJ log volumes significant values			
Variable	Estimate	Residual Std. Error	Significance
DJ_log_vol.l1	0,3675	0,2627 ***	
SPX_log_vol.l3	-0,1673	0,2627 **	
DJ_log_vol.l3	0,20735	0,2627 ***	
SPX_log_vol.l5	-0,14675	0,2627 **	
DJ_log_vol.l5	0,21354	0,2627 ***	
const	4,51269	0,2627 ***	
			R-squared
			0,4204
			F-statistic
			125,9

*** (p < 0)
 ** (p < 0.001)
 * (p < 0.01)
 . (p < 0.05)

S&P500 log prices differenced: DJ has a positive significant impact on lags 2,3,9 and a negative impact on lag 6. This means that past returns influence the S&P500. S&P500 is also negatively affected by its own third lag. R² explains only 5% of the total variance. Most of the information is actually delivered by DJ (as expected from our previous analysis).

DJ log prices differenced: DJ has a positive significant impact on lags 2 and 9 and a negative impact on lag 6; we have a negative impact of DJ prices in lag 6 and S&P500 has a negative impact on lag 3. R² explains only 5% of the total variance (similarly to S&P500).

S&P500 log volumes: Positive impact delivered by S&P500 on lag 1,2,6 and by DJ in lags 3,6. One negative for lag six of DJ. 30% of the variance is explained.

DJ log volumes: We have a significant impact of lag 1 by DJ, lag 3 from both S&P500 and DJ and again, lag 5 from both S&P500 and DJ. 40% of the variance is explained.

We can conclude by saying that log prices behave as Granger test suggested, even though it's not clear whether the relations highlighted by the VAR model are actually significant or spurious. It might be the case that price relation of Lag 6 and 9 is due to short term noise, while trading volumes, show very consistent and significant values and are more reasonable.

Conclusion:

This study applies several econometric techniques to investigate the relationship between two indexes, the S&P500 and the Dow Jones. We started our analysis by standardizing the two time series in order to be able to compare them and by plotting some graphs to get some initial insights. Log prices exhibit different residual behaviour compared to log volumes and required different treatments. As expected, initial tests suggested the need to detrend and first difference the log prices, while for the log trading volumes, trend had to be considered to proceed to modelling. Ljung Box test showed the need of further modelling; ARIMA(2,0,2) has been chosen as the best-fitting model according to AIC and BIC. For volatility analysis we have used a GARCH model which showed better explanatory power than the ARCH model according to the test we have run. After running several specifications for ARMA+GARCH, ARMA (2,2) + GARCH (1,1) revealed to be the most parsimonious according to the AIC/BIC criteria. Despite ARMA and GARCH models showed similar coefficients, significance and overall behaviour among log prices, the two time series are not cointegrated. VAR analysis shows that one series has explanatory power on the other and vice versa, depending on the modelling objective and time series of interest (and Lag component). Cointegration analysis also revealed that log trading volumes are more structured and predictable compared to log prices. The DJIA appears to play a leading role in information transmission, particularly when talking about log returns, while trading volumes offer a more stable ground for forecasting and interpretation.

Literature:

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